

ECON 695 Final Project

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1 Introduction

Understanding the determinants of restaurant revenue is important for both business owners and investors, as revenue outcomes influence decisions regarding expansion, staffing, and long-term viability. Restaurants differ substantially in their operational characteristics, location environments, and design choices, all of which may affect performance in complex and potentially nonlinear ways.

This paper evaluates how well restaurant revenue can be predicted using observable business characteristics and compares several predictive modeling approaches to assess their relative performance. The emphasis is on prediction rather than causal inference, with a particular focus on out-of-sample accuracy.

Research question: How well can restaurant revenue be predicted using observable restaurant characteristics, and which modeling approaches perform best in capturing these relationships?

The analysis compares linear and nonlinear models and evaluates their performance using a train–test framework, placing primary importance on predictive accuracy rather than interpretability alone.

1.1 Dataset

The dataset can be summarized as follows:

- Source: restaurant revenue dataset with 137 observations
- Outcome variable: annual restaurant revenue
- Predictors:
 - City group
 - Restaurant type
 - A set of numerical features (P1–P37)

The analysis uses a cross-sectional dataset of 137 restaurants containing information on annual revenue and a rich set of observable characteristics. These include categorical variables describing city size and restaurant type, as well as numerical indicators capturing operational and design features. The data are cross-sectional.

1.2 Method Overview

To evaluate predictive performance, the dataset is divided into training and testing samples. All models are estimated using the training data, and their performance is assessed on held-out observations that are not used during estimation. This train-test split ensures that model comparisons are based on out-of-sample predictive accuracy rather than in-sample fit. Several modeling approaches are considered, including ordinary least squares regression, regularized linear models such as ridge regression and the LASSO, and flexible machine learning methods such as random forests and gradient boosting. Using multiple models allows for a comparison between linear and nonlinear specifications, as well as an examination of the bias–variance tradeoff and the balance between interpretability and predictive power.

1.3 Main Findings and Contribution

The results reveal substantial differences in predictive performance across modeling approaches. Ordinary least squares performs poorly when evaluated out of sample, indicating that a simple linear specification is insufficient for accurate revenue prediction. Regularized regressions significantly improve performance by shrinking coefficients and reducing overfitting. Tree-based methods achieve the lowest prediction error overall, suggesting that nonlinearities and interactions play an important role in explaining revenue variation.

Variable importance measures differ across models, with restaurant age and specific operational characteristics emerging as more influential in nonlinear specifications than in linear regressions. Overall, this paper demonstrates how model choice affects both predictive accuracy and economic interpretation in applied settings with rich covariate information.

2 Background and Data Description

The dataset used in this study consists of information on 137 restaurants, originally collected to study revenue variation across different restaurant formats and locations. This dataset has been used previously in applied prediction settings, where the primary objective is to forecast restaurant revenue based on observable characteristics rather than to identify causal relationships. Prior work using this data has emphasized the difficulty of revenue prediction due to substantial heterogeneity across restaurants and the presence of many correlated covariates.

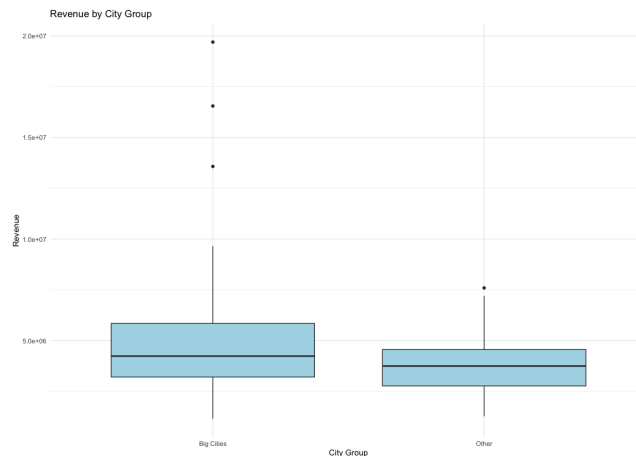


Figure 1: Restaurant Revenue by City Group

The data are cross-sectional, with each restaurant observed at a single point in time. The outcome variable is annual restaurant revenue. The covariates include a set of categorical variables describing the restaurant’s city group (large cities versus other locations) and restaurant type, as well as 37 numerical variables capturing operational, design, and business characteristics. In addition, the dataset includes the opening date of each restaurant, which allows for the construction of a measure of restaurant age. Because revenue is observed only once per restaurant and no time-varying outcomes are available, the analysis treats the data as cross-sectional rather than time-series.

The dataset contains 137 observations and 43 variables in total, including the outcome variable. No missing values are present in the data, eliminating the need for imputation or sample exclusion due to incomplete observations. Prior to estimation, several preprocessing steps are applied. Categorical variables are converted to factor variables to allow for appropriate treatment in regression and machine learning models. The opening date is converted to a date format and used to construct a continuous measure of restaurant age in years, defined as the difference between the most recent opening date in the sample and the restaurant’s opening date. The city name variable is excluded from the analysis to avoid issues related to high-dimensional categorical variables and to ensure consistent prediction in the testing sample.

Exploratory analysis reveals that restaurant revenue is highly right-skewed, motivating the use of a logarithmic transformation of revenue in several models. Summary statistics and visualizations further indicate substantial variation across restaurants in both revenue and operational characteristics. These features make the dataset well-suited for comparing linear and nonlinear predictive models and for examining how model choice affects out-of-sample performance.

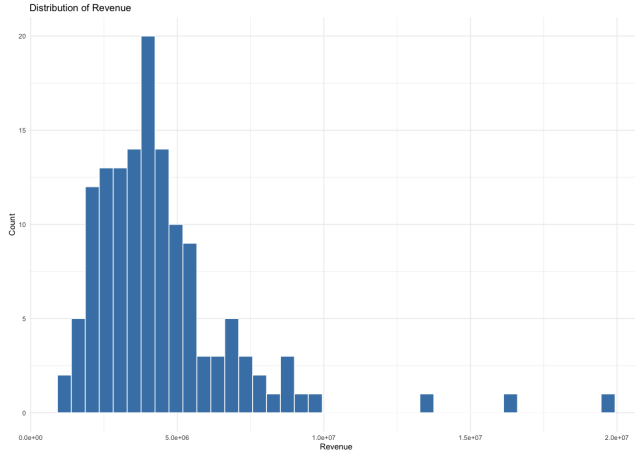


Figure 2: Distribution of Annual Restaurant Revenue

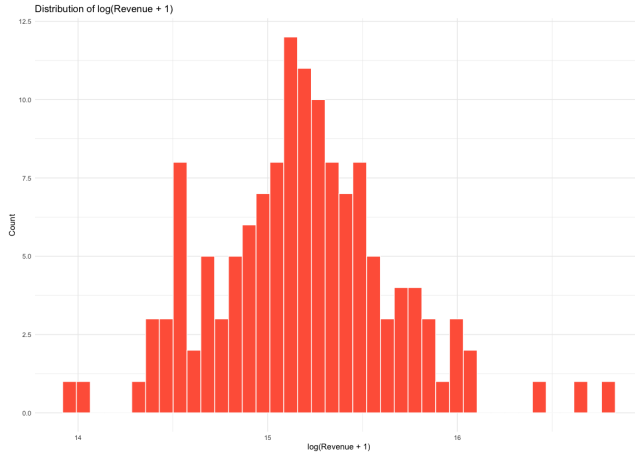


Figure 3: Distribution of Log-Transformed Revenue

Figures 2 and 3 illustrate the distribution of annual revenue and log-transformed revenue, highlighting the strong right-skewness of the outcome variable.

3 Model

The objective of this paper is to evaluate the ability of different statistical and machine learning models to predict restaurant revenue using observable characteristics. Because the goal is predictive accuracy rather than causal inference, the analysis focuses on out-of-sample performance and model comparison rather than coefficient interpretation alone.

As a baseline, ordinary least squares (OLS) regression is used to model revenue as a linear function of restaurant characteristics. This approach provides a familiar benchmark and allows for straightforward interpretation of coefficients. However, given the relatively small sample size and the large number of covariates, OLS may suffer from overfitting and poor out-of-sample performance. To address this concern, OLS is estimated both on raw revenue and on log-transformed revenue, reflecting the skewed distribution of the outcome variable.

Exploratory analysis reveals substantial correlation among many of the numerical predictors, particularly within the set of operational characteristics. This high degree of multicollinearity motivates the use of regularized regression methods and nonlinear models that are more robust to correlated inputs.

To improve predictive performance and mitigate overfitting, the analysis next considers regularized linear models. Ridge regression and the least absolute shrinkage and selection operator (LASSO) introduce penalties on coefficient magnitudes, shrinking less informative coefficients toward zero. Ridge regression reduces variance by shrinking all coefficients, while LASSO additionally performs variable selection by setting some coefficients exactly to zero. These models allow for a tradeoff between bias and variance and are well suited for settings with many correlated predictors.

Finally, the analysis employs nonlinear, tree-based methods: random forests and gradient boosting. These methods are designed to capture complex nonlinear relationships and interactions among covariates without requiring them to be specified in advance. Random forests aggregate predictions from many decision trees built on bootstrapped samples, reducing variance through averaging. Gradient boosting builds trees sequentially, with each tree focusing on improving the errors of the previous ones. While these models are less interpretable than linear regressions, they often achieve superior predictive performance when nonlinearities are present.

All models are estimated using the training sample and evaluated on a held-out testing sample. Predictive performance is assessed using root mean squared error (RMSE) and mean absolute error (MAE), computed both on the log-revenue scale and after transforming predictions back to the revenue scale. Comparing these models allows for an assessment of how linear versus nonlinear specifications perform in practice and highlights the role of model flexibility in revenue prediction.

4 Results

This section presents the out-of-sample predictive performance of the models described above. All models are estimated using the training sample and evaluated on a held-out testing sample. Model performance is assessed using root mean squared error (RMSE) and mean absolute error (MAE). For models estimated on log-transformed revenue, performance is reported both on the log scale and after transforming predictions back to the revenue scale.

4.1 Baseline Linear Models

Ordinary least squares regression performs poorly in out-of-sample prediction. When estimated on raw revenue, the OLS model yields a high prediction error and a negative out-of-sample R^2 , indicating that it performs worse than simply predicting the mean revenue in the test sample. Estimating OLS on log-transformed revenue improves in-sample fit but still results in large out-of-sample prediction error when predictions are converted back to the revenue scale. These findings suggest that a simple linear specification is insufficient to capture the relationship between restaurant characteristics and revenue. Figure 4

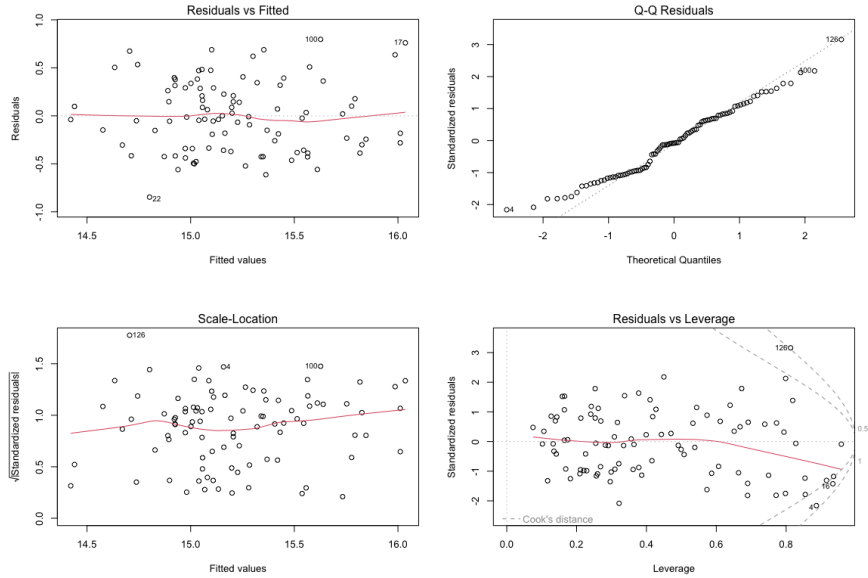


Figure 4: Random forest variable importance measures based on percentage increase in mean squared error and node purity. Restaurant age and several operational features are among the most influential predictors of revenue.

presents diagnostic plots for the log-revenue OLS model. The residual patterns indicate deviations from linearity and heteroskedasticity, while the Q-Q plot suggests departures from normality. These violations of classical OLS assumptions help explain the model's weak out-of-sample performance and motivate the use of more flexible approaches.

4.2 Regularized Regression Models

Regularization substantially improves predictive performance relative to OLS. Both ridge regression and the LASSO achieve significantly lower RMSE and MAE on the test sample. By shrinking coefficient estimates and reducing sensi-

tivity to noise in the data, these models mitigate overfitting and improve generalization to unseen observations. The similarity in performance between ridge and LASSO suggests that while variable selection is possible in this setting, much of the improvement comes from coefficient shrinkage rather than sparsity alone.

4.3 Tree-Based Models

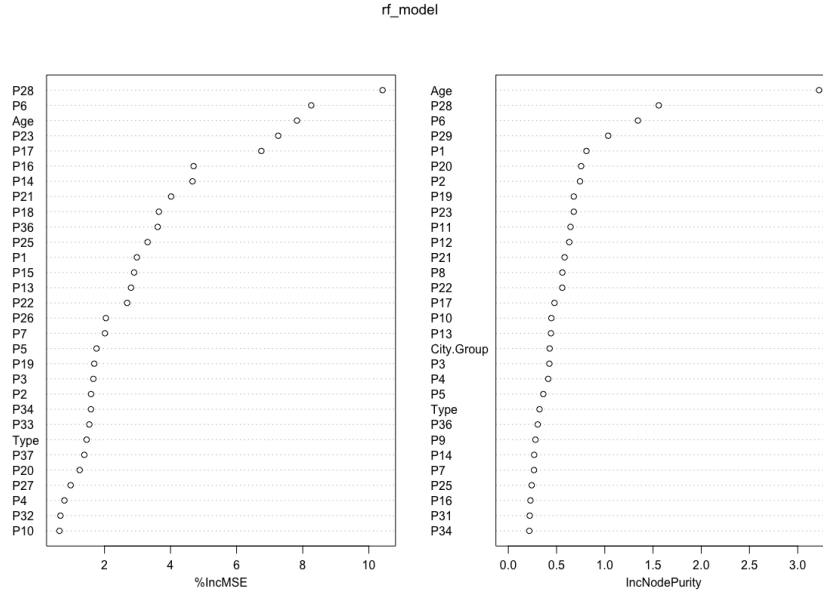


Figure 5: Random forest variable importance measures based on percentage increase in mean squared error and node purity. Restaurant age and several operational features are among the most influential predictors of revenue.

Tree-based models achieve the best predictive performance overall. Random forests produce the lowest out-of-sample RMSE and MAE among all models considered, followed closely by gradient boosting. These methods outperform linear and regularized regressions, indicating that nonlinearities and interactions among covariates play an important role in explaining revenue variation. Although these models sacrifice some interpretability, their superior predictive accuracy highlights the value of flexible modeling approaches in this context.

4.4 Variable Importance

Variable importance measures reveal notable differences across models. In the LASSO specification, only restaurant age retains a nonzero coefficient at the optimal penalty level, suggesting that many operational variables contribute

little to prediction in a linear framework. In contrast, gradient boosting assigns substantial importance to restaurant age, as well as to several operational characteristics such as measures related to capacity and layout. This divergence indicates that certain predictors influence revenue through nonlinear or interaction effects that linear models fail to capture.

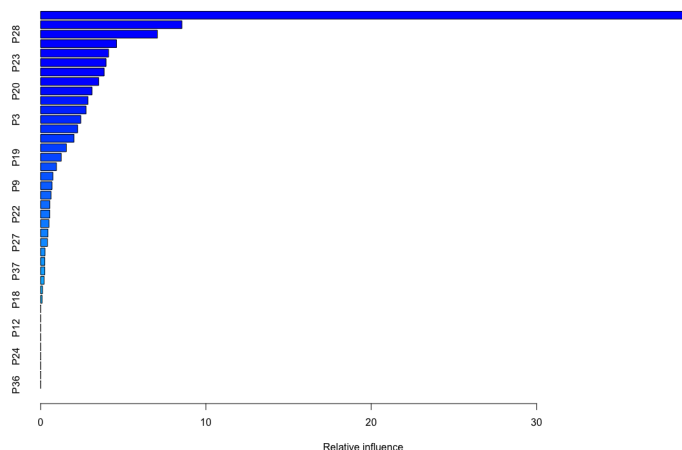


Figure 6: Variable Importance from Gradient Boosting Model

Overall, the results demonstrate that model choice has a significant impact on predictive accuracy. While linear models provide a useful benchmark, regularization and nonlinear methods offer substantial gains in performance. These findings underscore the importance of flexibility and careful model selection when predicting economic outcomes using rich covariate information.

5 Discussion

The results highlight the importance of model flexibility when predicting restaurant revenue using rich covariate information. The poor out-of-sample performance of ordinary least squares indicates that a linear specification is unable to capture key features of the data-generating process. In particular, the presence of many correlated predictors and potentially nonlinear relationships makes OLS especially susceptible to overfitting, leading to weak generalization performance on unseen data.

Regularized linear models offer a meaningful improvement by addressing the bias–variance tradeoff. Both ridge regression and the LASSO substantially reduce prediction error relative to OLS, demonstrating the value of coefficient shrinkage in high-dimensional settings. While the LASSO additionally performs

variable selection, the similarity in performance between ridge and LASSO suggests that most predictors contain some information, but their effects are difficult to isolate in a purely linear framework.

Tree-based methods outperform all linear models, providing evidence that nonlinearities and interactions among restaurant characteristics play an important role in determining revenue. Variables such as restaurant age, which appear weak or insignificant in linear regressions, become highly influential in nonlinear models. This pattern suggests that the effect of age on revenue is not constant but varies across restaurants, potentially interacting with operational scale or location characteristics. Linear models, which impose constant marginal effects, fail to capture these patterns.

Despite their strong predictive performance, tree-based models come with tradeoffs. Their complexity reduces interpretability relative to linear regressions, making it more difficult to attribute revenue differences to individual covariates. In applied economic settings, this tradeoff between interpretability and predictive accuracy is often central. While nonlinear models are preferable for prediction, simpler models may still be useful for descriptive analysis or policy interpretation.

Several limitations of the analysis should be noted. The sample size is relatively small, and the data are cross-sectional, limiting the ability to capture dynamic revenue patterns over time. Additionally, the analysis focuses on prediction rather than causal inference, and the estimated relationships should not be interpreted as causal effects. Future work could extend this analysis using panel data or explore hybrid approaches that combine interpretability with nonlinear flexibility.

Overall, the findings emphasize that predictive performance depends critically on model choice and that flexible machine learning methods can substantially outperform traditional linear approaches when rich feature sets are available.

6 Conclusion

This paper examines the problem of predicting restaurant revenue using observable business and location characteristics. Using a cross-sectional dataset and a train-test evaluation framework, the analysis compares the predictive performance of linear, regularized, and nonlinear machine learning models. The results show that simple linear regression performs poorly out of sample, while regularization and model flexibility substantially improve predictive accuracy.

Regularized regressions provide meaningful gains by reducing overfitting, but tree-based methods such as random forests and gradient boosting achieve the strongest performance overall. These models better capture nonlinear relationships and interactions among covariates, particularly with respect to restaurant age and operational characteristics. The findings highlight the importance of selecting models that align with the structure of the data when the objective is prediction rather than causal inference.

Overall, this study illustrates how model choice influences both predictive accuracy and interpretation in applied economic settings. When rich feature sets are available, flexible machine learning methods can offer significant advantages over traditional linear approaches, while the tradeoff between interpretability and predictive power remains central.

References

- [1] James, G., Witten, D., Hastie, T., & Tibshirani, R. (2023). *An Introduction to Statistical Learning with Applications in R* (2nd ed.). Springer.
- [2] Kaggle. (2015). *Restaurant Revenue Prediction*. Retrieved from <https://www.kaggle.com/competitions/restaurant-revenue-prediction/overview>

A Additional Details

A.1 Train-Test Split

To evaluate predictive performance, the dataset is randomly split into training and testing samples using a 70/30 split. All models are estimated exclusively on the training data, and performance metrics are computed using the held-out testing data. This approach ensures that results reflect out-of-sample predictive accuracy rather than in-sample fit.

A.2 Evaluation Metrics

Model performance is assessed using root mean squared error (RMSE) and mean absolute error (MAE). For models estimated on log-transformed revenue, metrics are reported both on the log scale and after transforming predictions back to the revenue scale. Reporting both sets of metrics allows for meaningful comparison across models while maintaining interpretability in terms of revenue units.

A.3 Model Tuning

Regularized regression models are estimated using cross-validation to select the penalty parameter that minimizes prediction error. For ridge regression and the LASSO, ten-fold cross-validation is used to determine the optimal regularization strength. Random forest models are estimated using 500 trees, with the number of variables considered at each split set to the square root of the number of predictors. Gradient boosting models are tuned using cross-validation to select the optimal number of trees.

A.4 Software

All analysis is conducted using R. The primary packages used include `ggplot2` for visualization, `glmnet` for regularized regression, `randomForest` for random forest estimation, and `gbm` for gradient boosting.

A.5 Correlation Structure of Predictors

Figure 7 displays the correlation matrix for the numerical predictors P1-P37. Many variables exhibit moderate to high pairwise correlations, indicating substantial multicollinearity. This correlation structure helps explain the poor out-of-sample performance of ordinary least squares and underscores the usefulness of regularization and tree-based methods, which are less sensitive to correlated covariates.

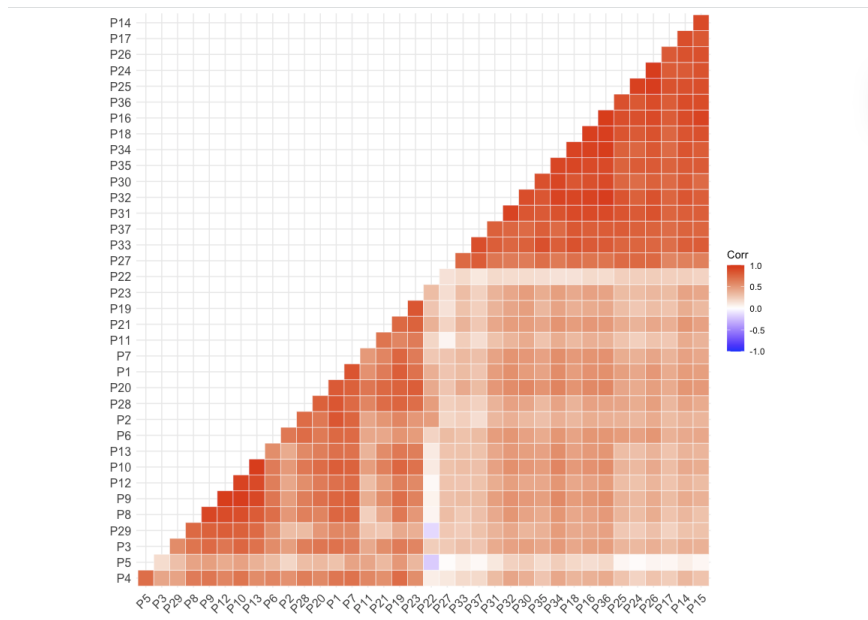


Figure 7: Correlation matrix for numerical predictors (P1–P37). Many variables exhibit moderate to high pairwise correlations, indicating substantial multicollinearity among operational characteristics.